FATIGUE FAILURE

Research references

Sources for the concepts explained in the film

01 Fatigue and crack growth

Roylance, D. (2001). **Fatigue**. MIT, Department of Materials Science and Engineering.

Used for repeated loading below yield, initiation and growth, fatigue-data scatter, and cyclic crack growth. See PDF pp. 1-4 and 6-7.

MIT OpenCourseWare - Fatigue (PDF)

02 S-N curves and material differences

Anand, L. and Parks, D. M. (2004). **Defect-Free Fatigue**. MIT 2.002 supplementary notes.

Used for stress amplitude, logarithmic stress-life plots, and the distinction between steel endurance limits and aluminum-alloy trends. See PDF pp. 14 and 17-19.

MIT OpenCourseWare - 2.002 lecture notes (PDF)

03 Stress intensity and fracture resistance

Roylance, D. (2001). **Introduction to Fracture Mechanics**. MIT.

Used for Mode I stress intensity, crack geometry and loading, toughness, and the limits of linear-elastic fracture mechanics.

MIT OpenCourseWare - Introduction to Fracture Mechanics (PDF)

04 Meaning and validity of K_Ic

NASA (2023). **NASA-HDBK-5010, Volume 1, Revision A**: Fracture Control Implementation Handbook for Spaceflight Hardware.

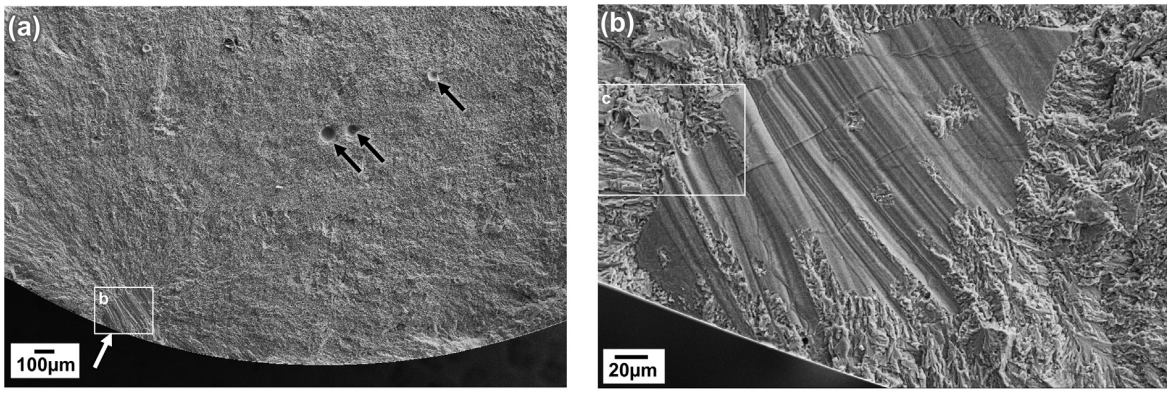
Used for the definition of linear-elastic, plane-strain fracture toughness and the need to satisfy validity conditions. See printed/PDF p. 25.

NASA Standards - Fracture Control Implementation Handbook (PDF)

Real photographs & credits

The scientific photographs are authentic; the motion diagrams are original schematics.

05 Real fatigue-fractured titanium specimen



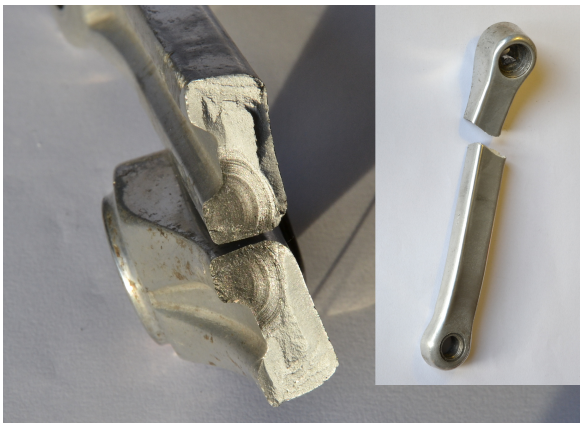
Credit: National Institute of Standards and Technology (NIST). Uploaded 7 February 2017. SEM images of additive-manufactured Ti-6Al-4V fatigue fracture surfaces.

The original white arrow identifies initiation at a lack-of-fusion defect. Black arrows indicate pores. Used at 00:23-00:33; resized with original arrows and scale bars retained.

NIST image page and description

NIST reuse policy: unmarked site information may be copied with appropriate credit. No separate copyright restriction is shown on this image page.

06 Real broken bicycle crank



Photographer: Oxensepp
File: 0x-kurbel-def.jpg
Date: 4 January 2014
Source: own work, Wikimedia Commons

Used at 02:31-02:40. The source description identifies slow fatigue growth and sudden fracture. The video resizes the image and adds explanatory arrows and labels.

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Illustrative data & production notes

How to interpret the numbers, drawings and timing

The S-N numbers are invented teaching examples

No numerical S-N point, crack dimension, material toughness or service-life estimate is claimed to be a measurement of a bicycle crank. All scatter points are synthetic.

Example	Stress amplitude	Cycles to failure
Higher amplitude	300 MPa	10,000
Lower amplitude	180 MPa	1,000,000

The renderer uses S = 540 - 60 log10(N), with S in MPa, only over N = 10^3 to 10^7. It is a convenient illustrative curve, not a fitted material model. Comparisons assume the same material, mean stress, surface and environment.

What the fracture animation assumes

K_I = Y * sigma * sqrt(pi * a). Y is the geometry factor; sigma is the applied tensile stress; a is the crack-length parameter. The paired animations hold Y constant: doubling stress doubles K_I, and increasing a by four doubles K_I. In real parts, Y can vary with crack geometry.

K_I describes loading at a crack tip. K_Ic describes material resistance for a valid Mode I, linear-elastic, plane-strain test condition. Maximum K_I reaching K_Ic is the rapid-fracture criterion shown under those assumptions. A real crank may require another fracture-resistance measure if those assumptions fail.

Fatigue growth can occur below K_Ic. Its rate also depends on the cyclic stress-intensity range, load ratio, environment and other conditions. K_Ic is not a fatigue-growth threshold.

Exact schedule and reproducibility

00:00-02:15: concepts, including the title and outline (135 s, 75%).
02:15-03:00: bicycle application, care and source credits (45 s, 25%).
Media: 1920 x 1080, 24 fps, H.264 video and AAC English speech; English captions burned into the picture. A separate SRT is included.

Source package: Python rendering and audio scripts, storyboard, real-photo assets, fonts and licenses, word timings, reusable narration, reference links, and the assignment prompt. Voice: en-US-AriaNeural via edge-tts. Bicycle art and diagrams are drawn by code.